

OBJECTIVES

- Students to start working on the UI using React for the DMS

INDIVIDUAL EXERCISES

Task / Exercise C1 [30 Minutes] (Hello World UI in React)

1. Install Node.js then run `npm create vite@latest dms-ui -- --template react`
2. Run `npm install` & `npm run dev` – confirm the app opens at `localhost:5173`
3. Replace the default `App.jsx` with a component that renders an `<h1> "Document Management System"` and your student name
4. Add a simple `AppHeader` component with the app name and a placeholder avatar icon
5. Apply minimal CSS: a top navbar bar with background color, centered content area with max-width 1200px

Task / Exercise C2 [30 Minutes] (Login Screen to Take to Hello Dashboard)

You need to continue from the previous project, to create a two-screen app : a login page and a basic dashboard – wired together with React Router.

1. Install `npm install react-router-dom` and set up `BrowserRouter` with two routes: `/` (Login) and `/dashboard`
2. Build a `LoginPage` component with two inputs (email + password), a submit button
3. On form submit, validate that neither field is empty, then call `useNavigate()` to redirect to `/dashboard`
4. Build a `DashboardPage` that shows "Welcome back, [username]" using state passed via `location.state`
5. Add a "Logout" button on the dashboard that navigates back to `/` and clears the state
6. Protect the `/dashboard` route – redirect to `/` if no user is logged in

Task / Exercise C3 [30 Minutes] (State Management + Building the DMS)

Students should start to create the different parts of the Document Management system including the dashboard for IT administrators and end-users via exploring :

- Integration of CSS libraries (Bootstrap, Tailwind...)
- Use of State Management libraries (starting with Context+Reducers).
- Use Fake API calls to test their solutions.

NEXT DELIVERABLE : 18 APRIL 2026 : Front-end UI : Individual Submission

Students need to develop a React app using Vite, React Router, Context + `useReducer` for state, and `json-server` as the fake API. No integration with Backend

- Login page with role detection (user vs admin) → redirects to the correct layout
- End-user view:
 - document list with search + filters + pagination + document upload modal, document detail with comments box from different users.
 - Screen to view Document detail including Meta data + previous versions + option to create a new version.
- Admin view:
 - users management (create / edit / suspend) List View with pagination + Multi-select Actions (Including Suspend, Export) + Import Wizard from CSV.
 - department management including assigning users to departments
 - Category management
- Students should create at least two automated E2E tests for their UI.
- Extra Features (As students will be using LLMs for the above, we will be marking mostly based on the extra innovative features being developed).